Byte Barnaby: Autonomous Multimodal 3D News Broadcaster

Project Background

The landscape of digital media and content creation is shifting toward automation, yet producing high-quality, fully animated 3D video content remains bottlenecked by expensive diffusion models, and reliance on closed-source, cloud-dependent APIs. Current solutions often suffer from high latency, rigid presentation formats, and massive computing costs.

Byte Barnaby was developed to bridge this gap by introducing a fully autonomous, local-first multimodal AI pipeline. By leveraging the computational power of AMD Radeon GPUs, this project transforms keywords given by users into a fully synthesized 3D WebGLnews broadcast. It proves that end-to-end multimodal content creation - combining Large Language Models, Zero-Shot Text-to-Speech, and acoustic forced alignment can be orchestrated seamlessly on local bare-metal hardware.

Target Users & Application Scenarios

This system is designed to democratize high-end production capabilities for creators and enterprises alike.

Target Users:

- **Independent Media & News Outlets**: Organizations looking to run 24/7 automated news broadcasts.
- **VTubers & Livestreamers:** Creators needing autonomous AI agents capable of reacting to events and interacting with audiences dynamically.
- **Game Developers:** Developers requiring lightweight, dynamically generated NPC dialogue and lip-syncing streamed directly into web engines.

Application Scenarios:

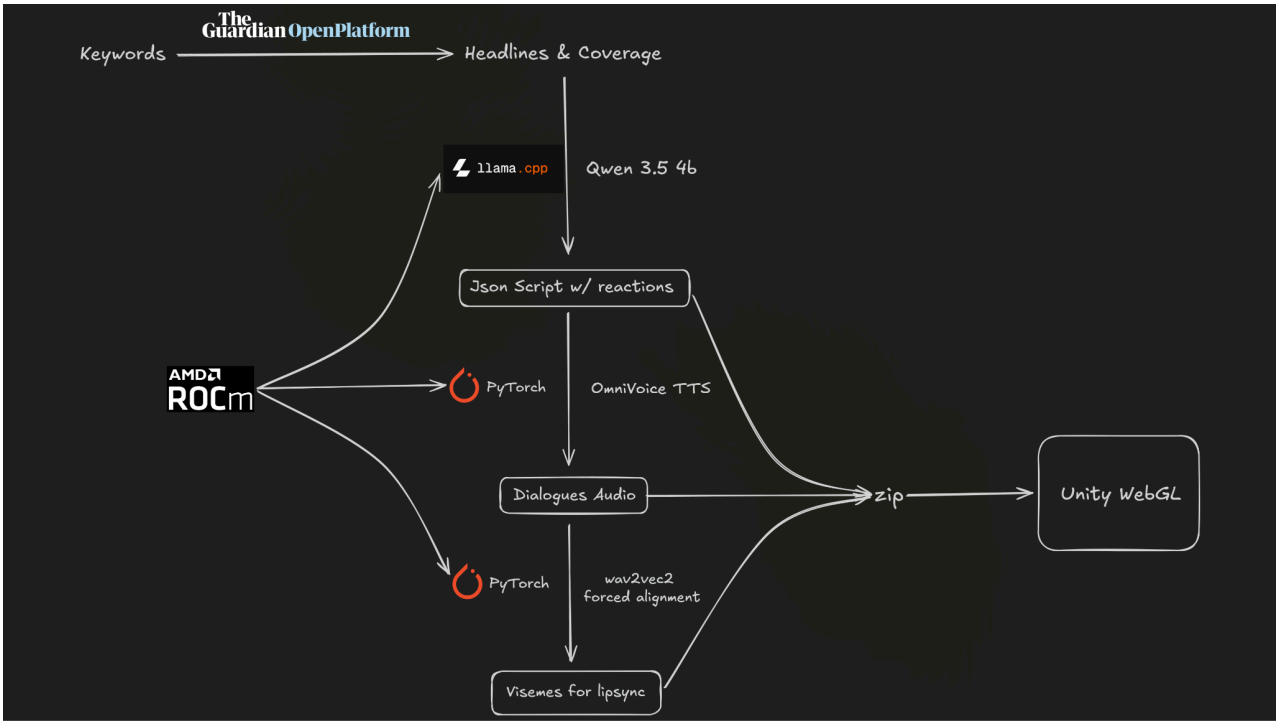
- **Automated 24/7 Broadcast Stations:** Byte Barnaby can continuously keep broadcasting and delivering news autonomously without human intervention,

anyone using it can just decide which keywords the broadcast should focus on.

- **Accessible Content Delivery:** Converting text-heavy articles into highly engaging, expressing audio-visual formats for visually impaired or neurodivergent audiences.

System Architecture

The system is built on a decoupled architecture, ensuring that heavy AI inference remains on the ROCm backend while delivering a lightweight, optimized experience to the user via browser.



Workflow Pipeline:

1. **Data Ingestion**: News headlines and coverage are fetched dynamically via The Guardian OpenPlatform API based on targeted keywords.
2. **Script & Reaction Generation:** The raw news data is processed by an LLM (Qwen 3.5 4b) running via llama.cpp on ROCm backend, which outputs a structured JSON script complete with dialogues lines, and emotional reactions to be used later by facial blendshapes.
3. **Audio Generation:** OmniVoice TTS is used via PyTorch ROCm to generate audio dialogues.

4. **Forced Alignment:** The generated audio and transcript are simultaneously processed by a wav2vec2 model, also via PyTorch, to calculate frame-accurate visemes for mouth movements.
5. **Payload Streaming:** The JSON script, audio files and viseme data are compressed into a .zip payload, the Unity build receives it, and starts broadcasting the show after decompressing.

Model & Algorithm Introduction

The system utilizes a stack of highly optimized open-weights models:

- **Qwen 3.5 (4B)**: Served via llama.cpp, this highly efficient LLM acts as the director. It excels at contextual reasoning, converting dry news into engaging, emotionally rich broadcaster script while adhering strictly to JSON output formatting.
- **OmniVoice TTS (0.6B):** A powerful Zero-Shot TTS model. It handles the generation of the anchor's voice, with highly expressive audio.
- **Wav2Vec2 (Forced Alignment):** A state-of-the-art acoustic model used to map spoken audio directly to text transcripts. The algorithm calculates precise timestamps for individual phonemes, which are then translated into blendshape weights (visemes) for the Unity 3D rig, ensuring flawless, mathematically aligned lip-syncing without manual animation.

Adaptation Description for AMD Radeon GPU/ ROCm

The entire AI inference backend of Byte Barnaby is custom-tailored to maximize the performance of AMD Radeon GPUs using the ROCm ecosystem.

- **Native LLM ROCm Execution:** Instead of relying on generalized Vulkan execution, the orchestration layer utilizes the official, pre-compiled llama.cpp ROCm 7.2 release binaries. This ensures that Qwen 3.5 4B token generation operates directly on bare-metal AMD silicon, leveraging native ROCm runtime libraries for maximum token-per-second output.

- **Native PyTorch ROCm Integration:** The audio pipeline (OmniVoice TTS and Wav2Vec2 Forced Alignment) runs natively on PyTorch configured for ROCm 7.2. Deep learning tensor operations and matrix multiplications heavily utilize AMD's MIOpen (HIP) primitives. As demonstrated in the terminal logs, the models successfully allocate and evaluate directly through MIOpen.
- **Compute Saturation:** System monitoring via rocm-smi confirmed that the pipeline effectively pushes the Radeon hardware to 100% GPU utilization proving efficient, unbottlenecked hardware adaptation for complex multimodal tasks.

